

GIT PULL

Git pull is used to fetch and update changes from a remote repository to the local repository.

GIT MERGE

Git merge is used to combine changes from one branch into another branch.

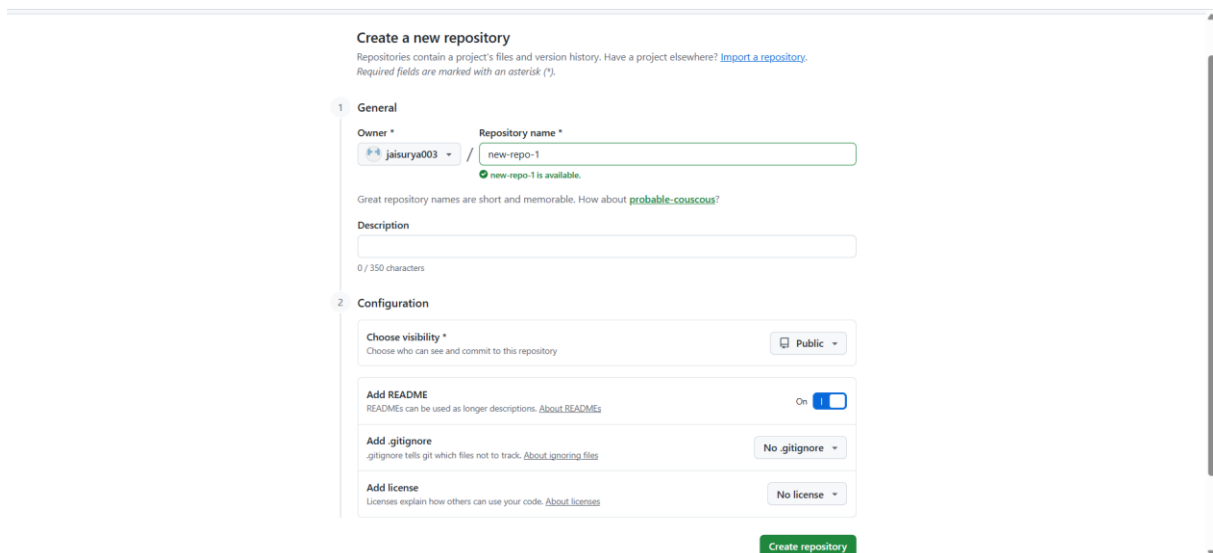
GIT PULL REQUEST

Git Pull Request Used to request merging changes from one branch into another branch in a GitHub repository.

HANDS-ON FOR GIT PULL AND GIT MERGE

Screenshot: Create New Repository

- Open GitHub
- Click New repository
- Enter repository name
- Enable README file
- Click Create repository

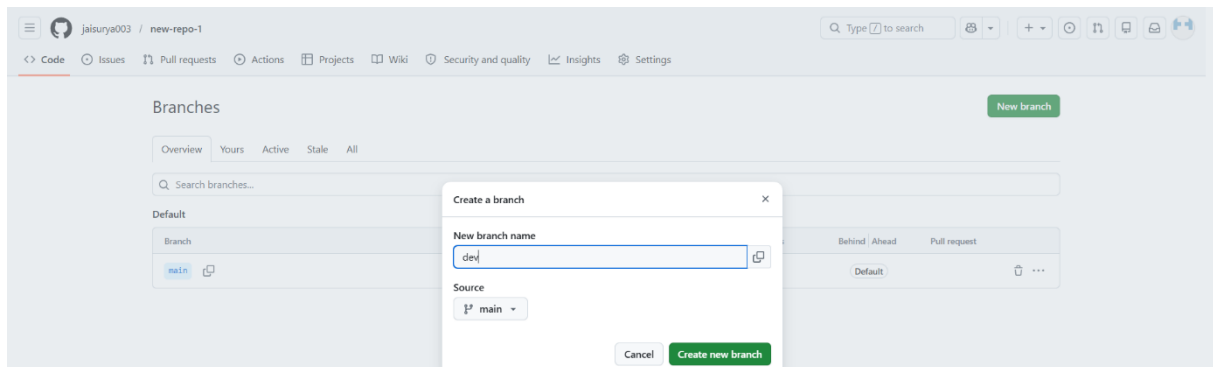


The screenshot shows the 'Create a new repository' page on GitHub. The page is divided into two sections: 'General' and 'Configuration'. In the 'General' section, the 'Owner' is set to 'jaisurya003' and the 'Repository name' is 'new-repo-1'. A green checkmark indicates that 'new-repo-1' is available. Below this, there is a 'Description' field with a character count of '0 / 350 characters'. The 'Configuration' section includes 'Choose visibility' set to 'Public', 'Add README' with a toggle switch turned 'On', 'Add .gitignore' set to 'No .gitignore', and 'Add license' set to 'No license'. A green 'Create repository' button is at the bottom right.

Screenshot: Create New Branch

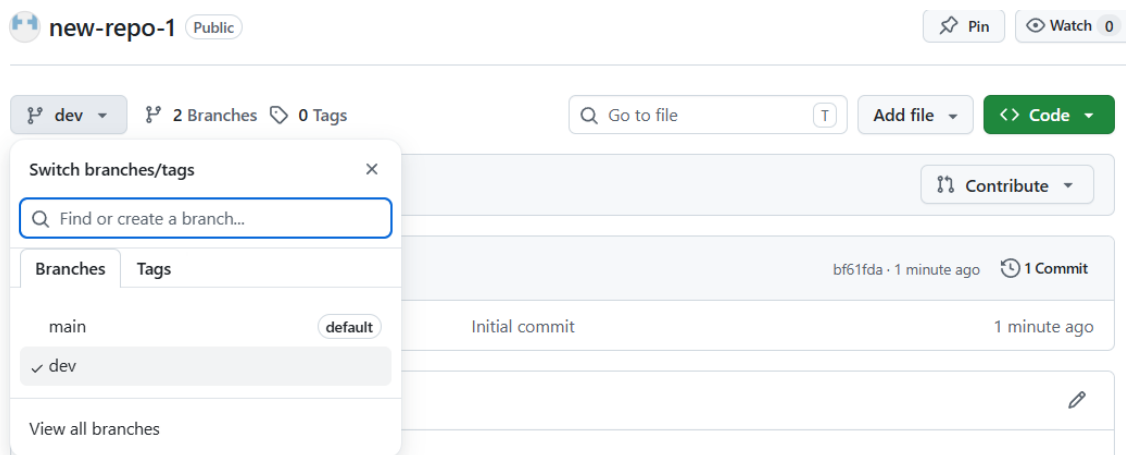
- Repository created successfully
- Click Branch
- Click New branch

- Enter branch name
- Click Create branch



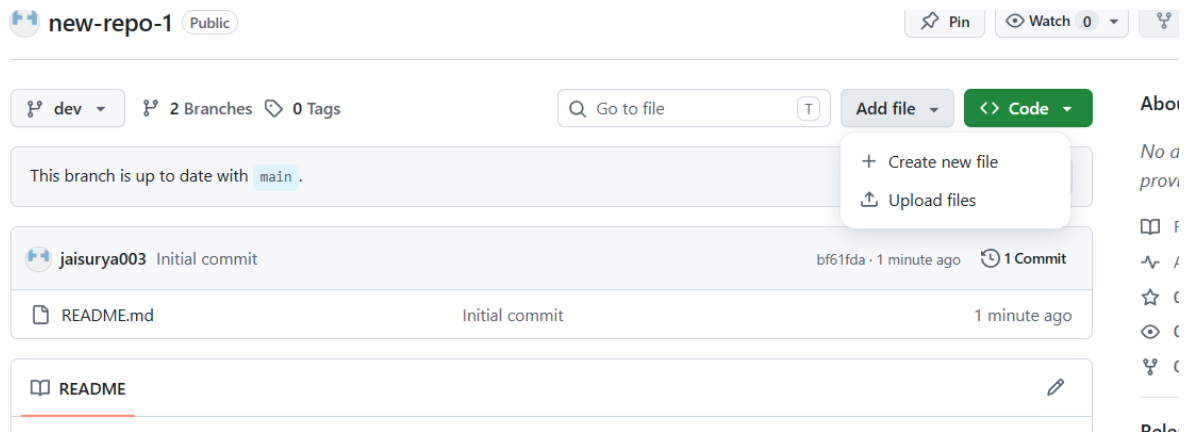
Screenshot: Branch Created Successfully

- New branch created successfully
- Branch added to the repository



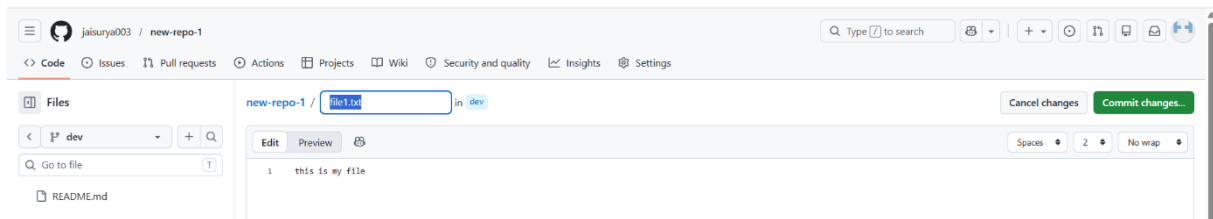
Screenshot: Create File in New Branch

- Switch to newly created branch
- Click Add file
- Click Create new file



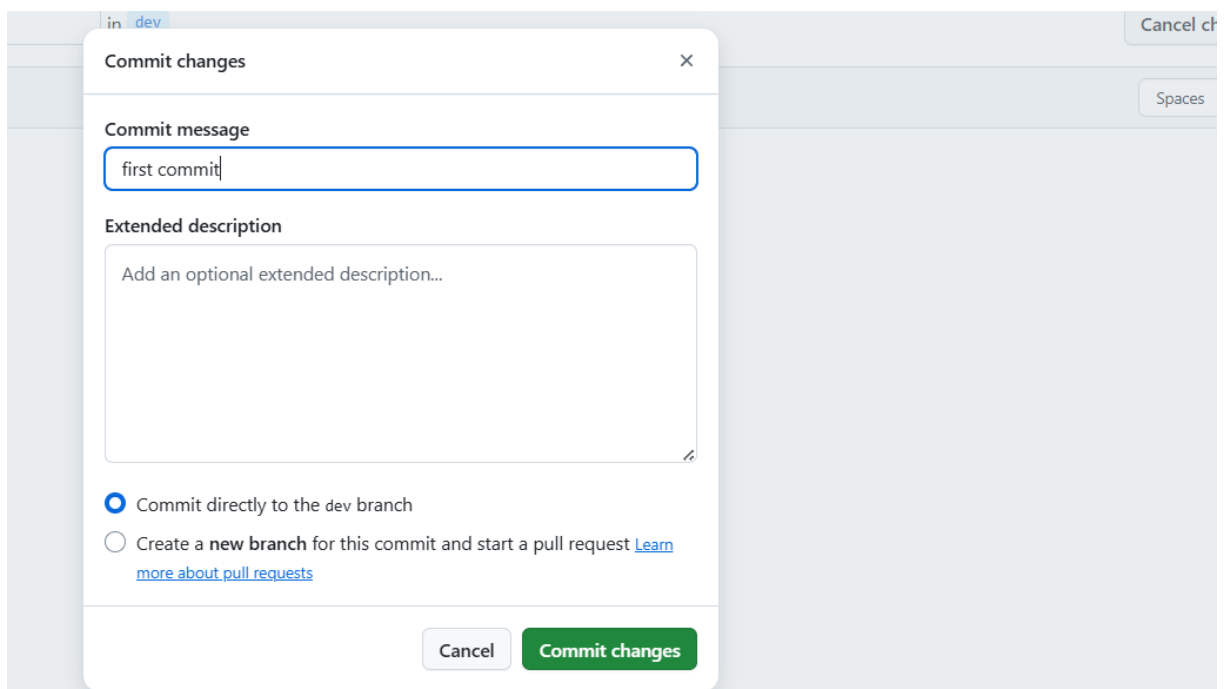
Screenshot: Commit File in Branch

- Enter file name and file content
- Click Commit changes



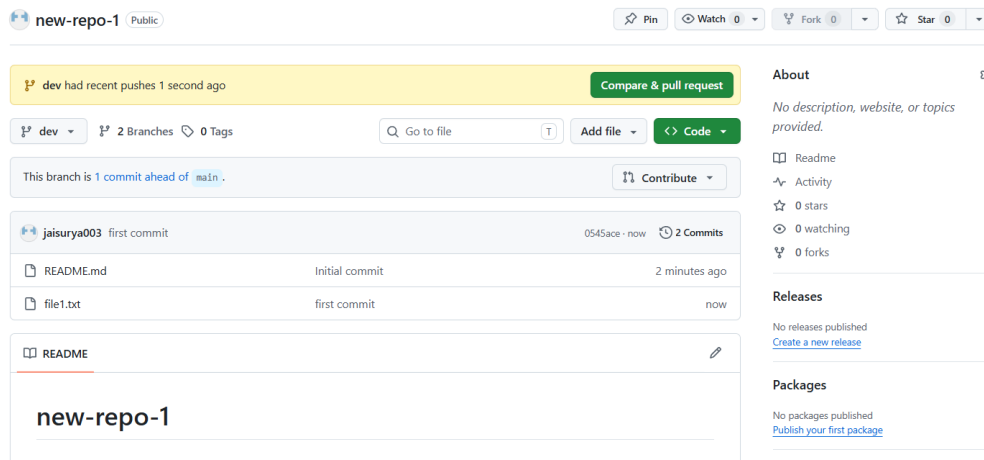
Screenshot: Add Commit Message

- Enter commit message
- Click Commit changes



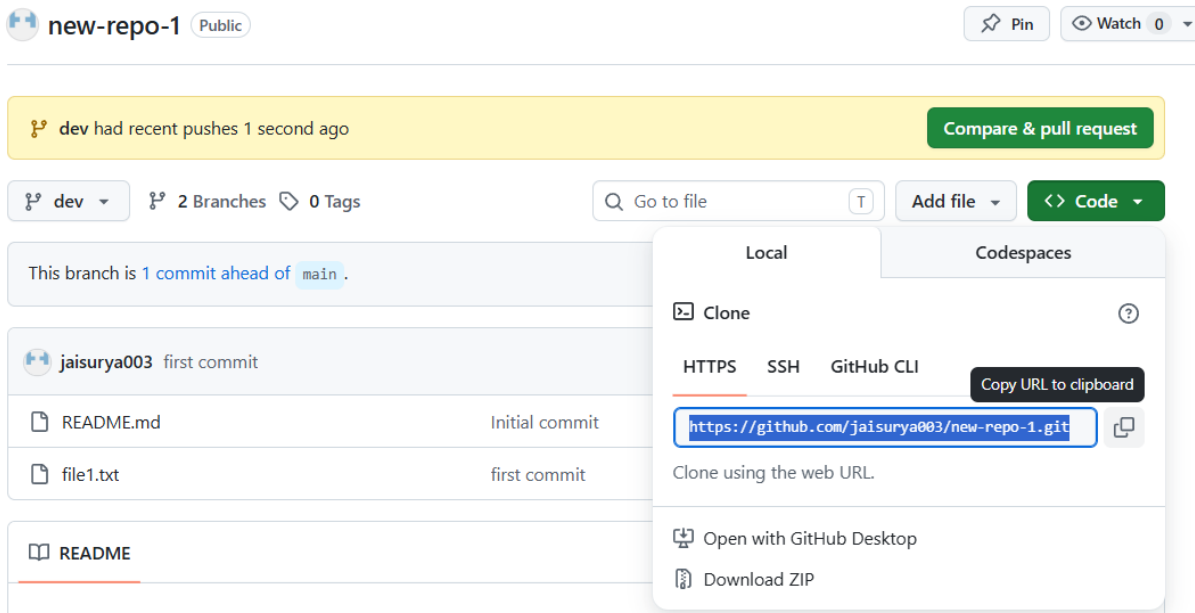
Screenshot: File Created in New Branch

- File created successfully in new branch
- Changes committed to the branch



Screenshot: Copy Repository Link

- Click Code
- Copy HTTPS repository link



Screenshot: Clone Repository and Switch Branch

- Open Git Bash
- Execute `cd Downloads` (to move into Downloads folder)
- Execute `git clone "repository link"` (to clone the GitHub repository)
- Execute `cd repository-name` (to move into repository folder)
- Execute `git switch new-branch` (to switch to the new branch)

```
MINGW64~/Users/JAISURYA/downloads/my-repo
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~
$ cd downloads
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads
$ git clone https://github.com/jaisurya003/my-repo.git
Cloning into 'my-repo'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads
$ cd my-repo
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (main)
$ git switch develop
branch 'develop' set up to track 'origin/develop'.
Switched to a new branch 'develop'
```

Screenshot: Pull Changes from Branch

- Execute `git pull origin dev` (to download latest changes from dev branch)
- Execute `ls` (to view repository files)
- Execute `git log --oneline` (to view commit history in one line format)

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (develop)
$ git pull origin develop
From https://github.com/jaisurya003/my-repo
 * branch      develop    -> FETCH_HEAD
Already up to date.
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (develop)
$ ls
README.md  newfile.py
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (develop)
$ git log --oneline
06139cc (HEAD -> develop, origin/develop) first commit
6545963 (origin/main, origin/HEAD, main) Initial commit
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (develop)
$
```

Screenshot: Merge Dev Branch into Main Branch

- Execute `git switch main` (to switch to main branch)
- Execute `ls` (to view repository files)
- Execute `git merge dev` (to merge dev branch into main branch)
- Execute `ls` (to verify merged files)
- Execute `git log --oneline` (to view updated commit history)
- Execute `git push origin main` (to push merged changes to GitHub)

```

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (develop)
$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (main)
$ ls
README.md

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (main)
$ git merge develop
Updating 6545963..06139cc
Fast-forward
 newfile.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 newfile.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (main)
$ ls
README.md newfile.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (main)
$ git log --oneline
06139cc (HEAD -> main, origin/develop, develop) first commit
6545963 (origin/main, origin/HEAD) Initial commit

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/my-repo (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/jaisurya003/my-repo.git
 6545963..06139cc  main -> main

```

Screenshot: File Merged into Main Branch

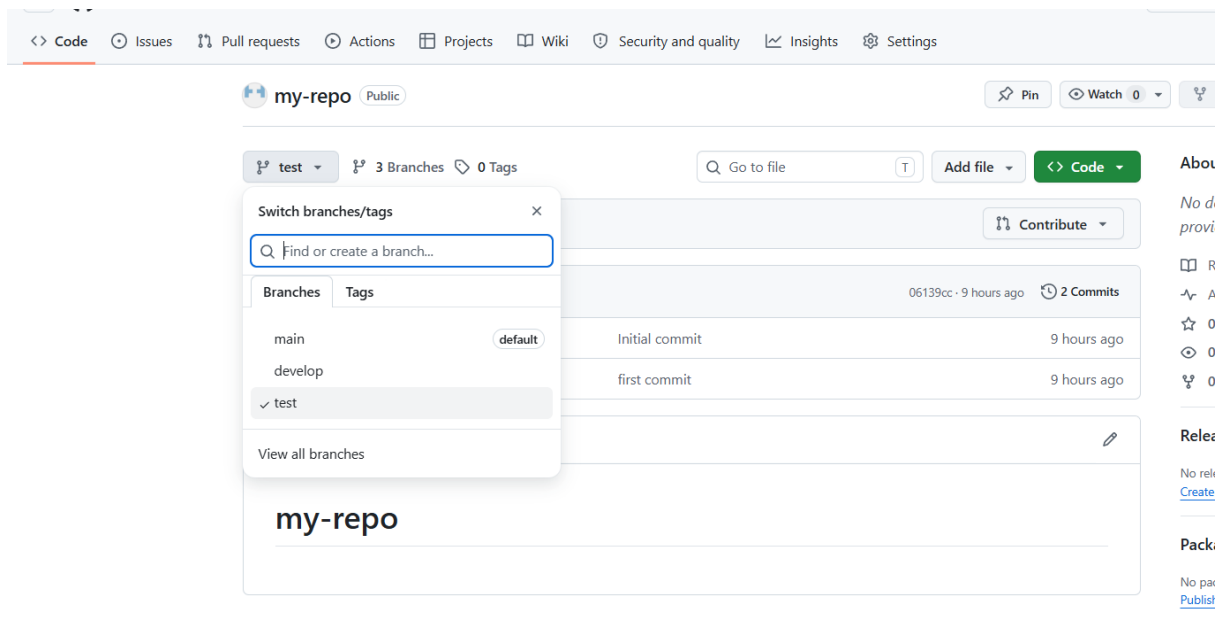
- File successfully merged into main branch
- Updated changes pushed to GitHub repository

The screenshot shows the GitHub interface for a repository named 'my-repo' by user 'jaisurya003'. The repository is public and has 2 branches and 0 tags. The 'main' branch is selected. The commit history shows two commits: 'first commit' by 'jaisurya003' at 06139cc, 9 minutes ago, and 'Initial commit' at 6545963. The file changes section shows 'README.md' (Initial commit, 10 minutes ago) and 'newfile.py' (first commit, 9 minutes ago). The README file is visible, showing the title 'my-repo'.

GIT PULL REQUEST

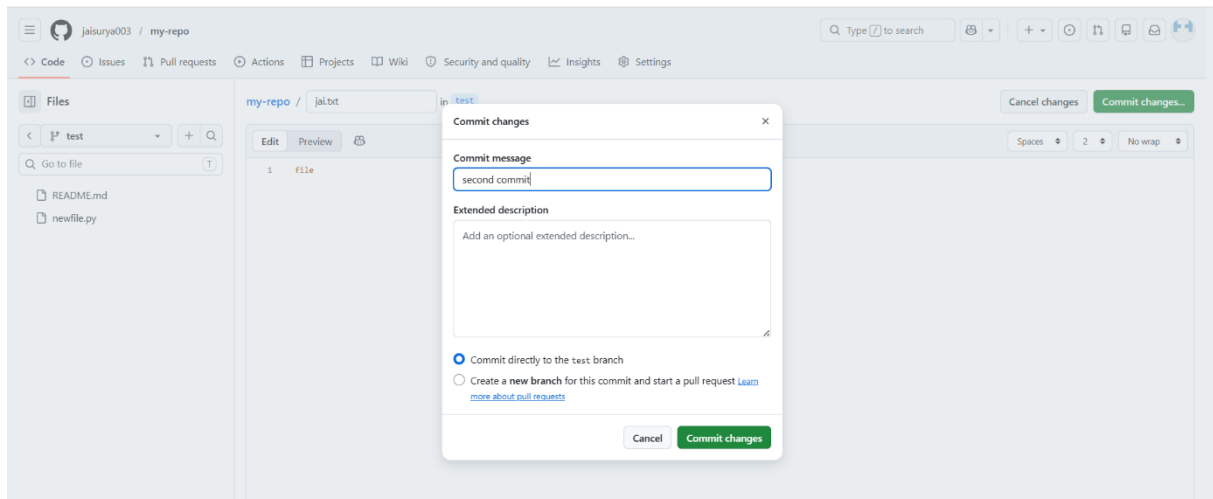
Screenshot: Create and Switch to Test Branch

- Open GitHub repository
- Click Branch
- Click New branch
- Enter branch name as test
- Click Create branch
- Switch to test branch



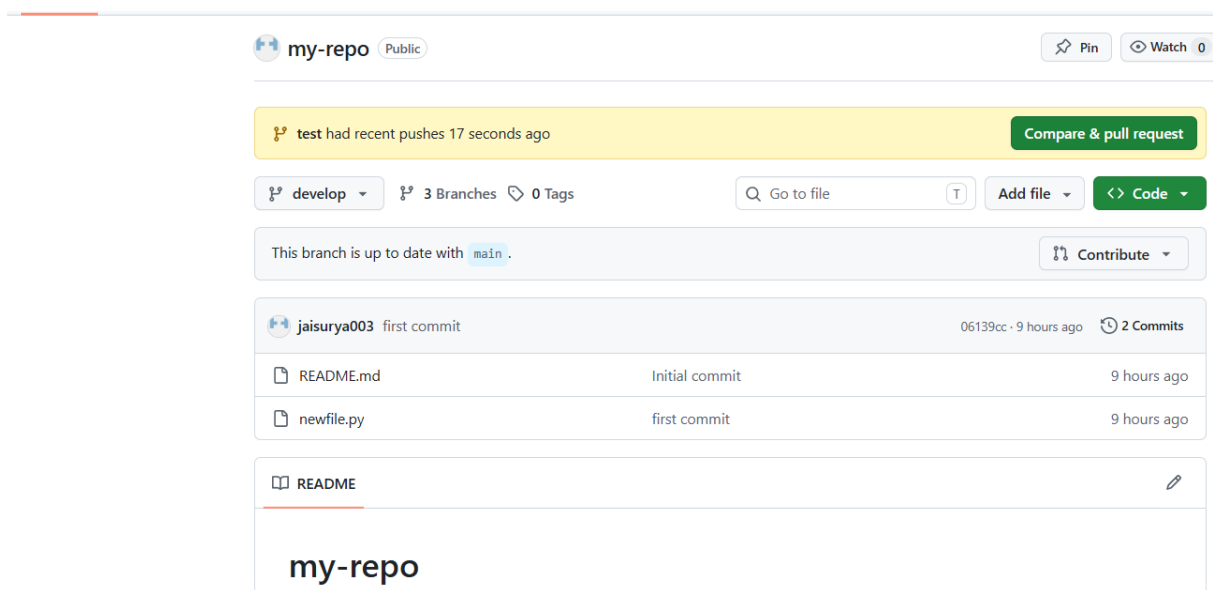
Screenshot: Create File in Test Branch

- Click Add file
- Click Create new file
- Enter file name and file content
- Click Commit changes
- Enter commit message
- Click Commit changes



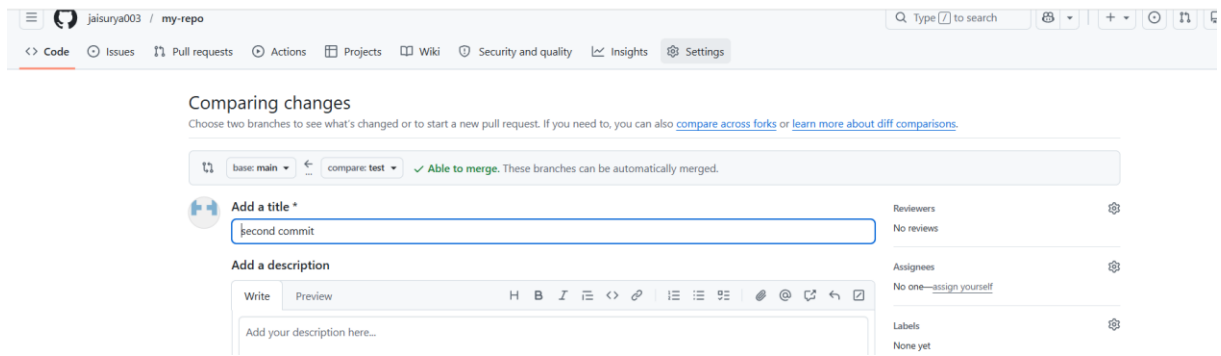
Screenshot: Create Pull Request from Test Branch

- Click Compare & pull request
- Review branch changes
- Click Create pull request



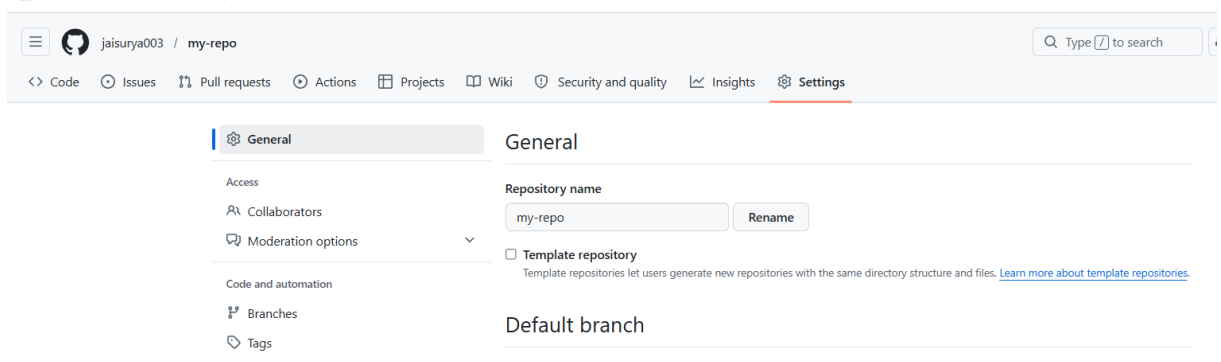
Screenshot: Open Repository Settings

- Go repository
- Click repository Settings icon



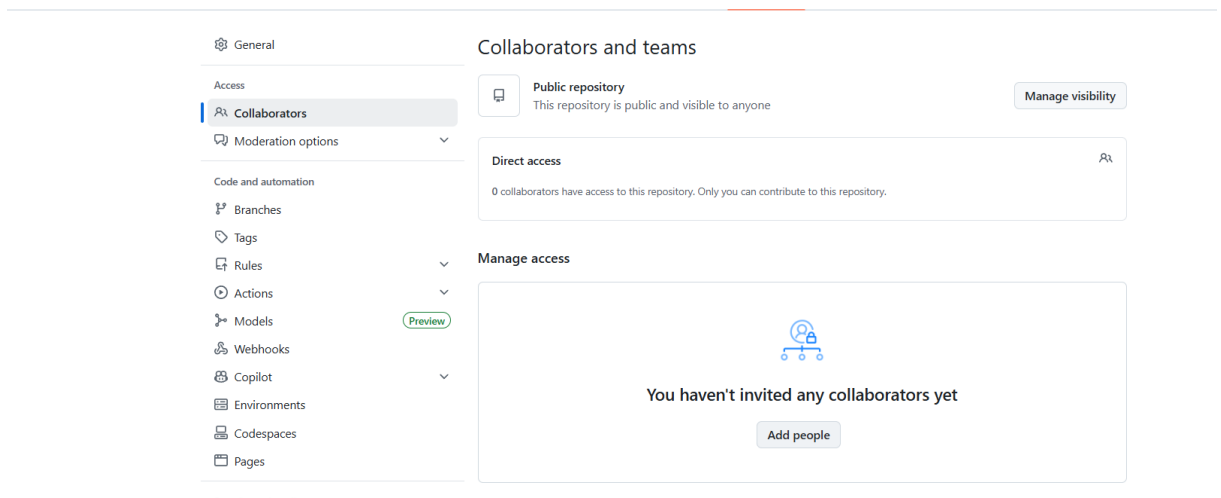
Screenshot: Open Collaborators Section

- Click Collaborators
- Open collaborators management section



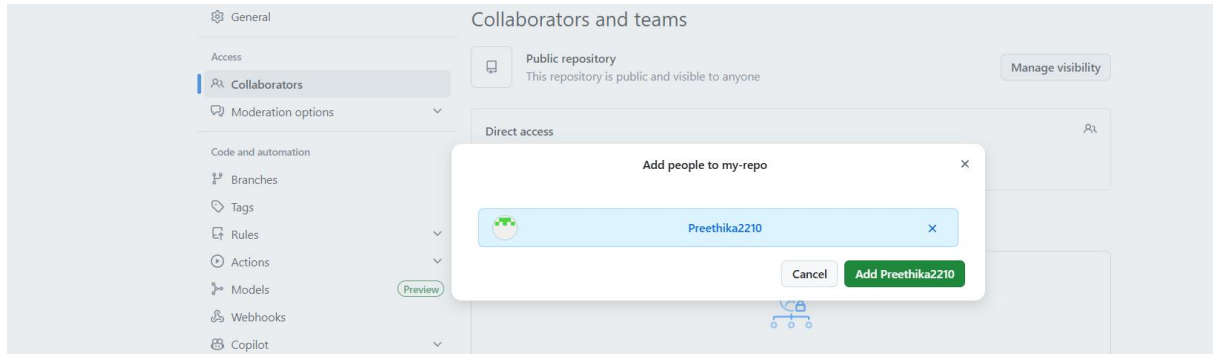
Screenshot: Add Collaborator

- Click Add people
- Open add collaborator section



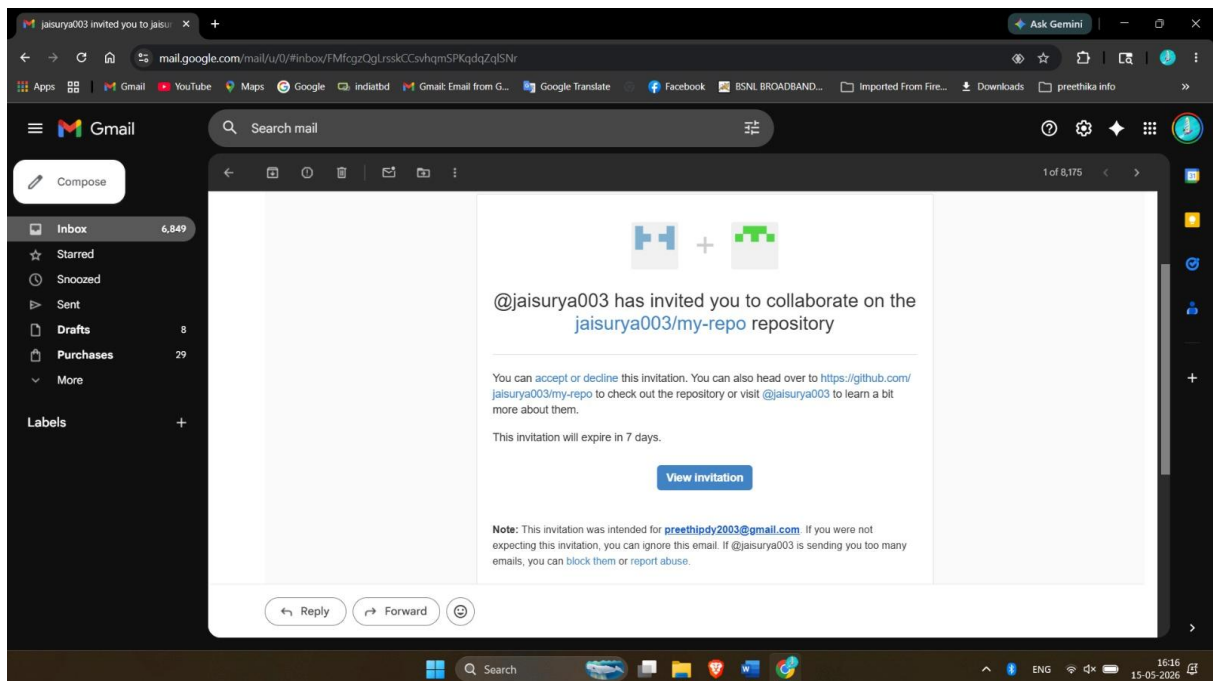
Screenshot: Search and Add Collaborator

- Search for collaborator username
- Select the user
- Click Add collaborator



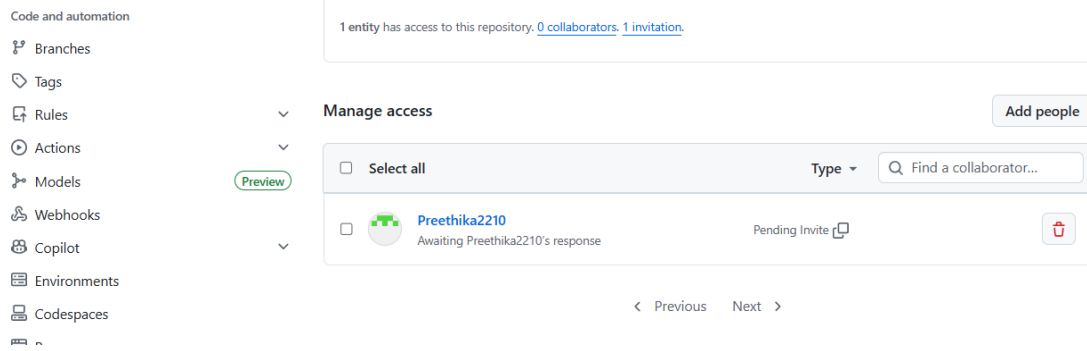
Screenshot: Collaborator Invitation Sent

- Collaborator invitation sent successfully
- Collaborator received invitation email



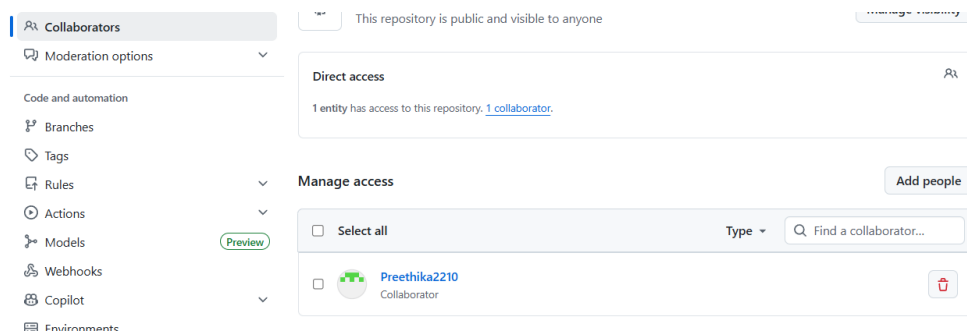
Screenshot: Pending Collaborator Invitation

- Collaborator invitation is in pending status
- Waiting for collaborator acceptance



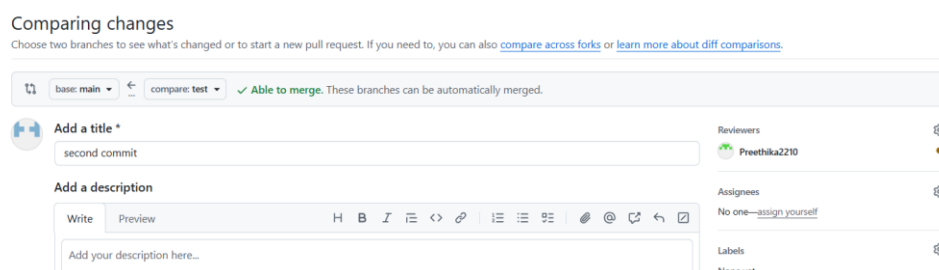
Screenshot: Collaborator Invitation Accepted

- Collaborator accepted the invitation
- Collaborator added successfully to the repository



Screenshot: Add Reviewer in Repository Settings

- Go to Reviewers settings
- Click Add reviewer
- Select reviewer for pull request approval

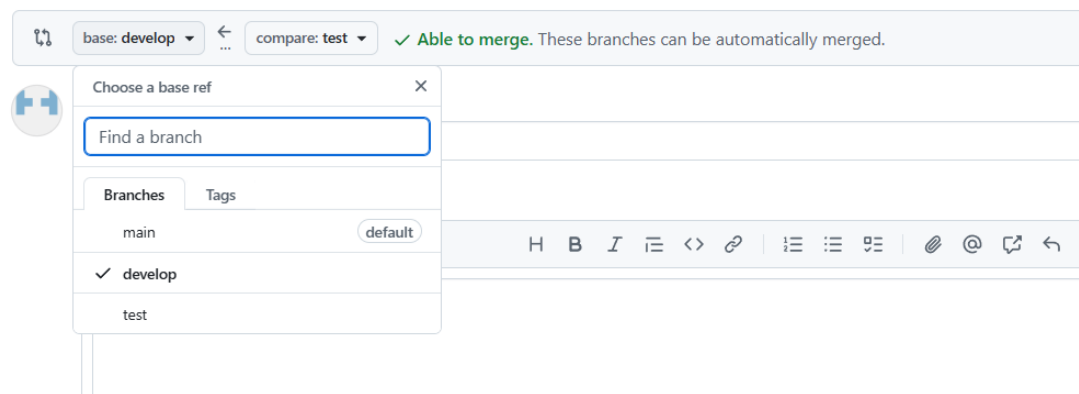


Screenshot: Compare Branches for Pull Request

- Select base branch as develop
- Select compare branch as test
- Review branch changes before pull request creation

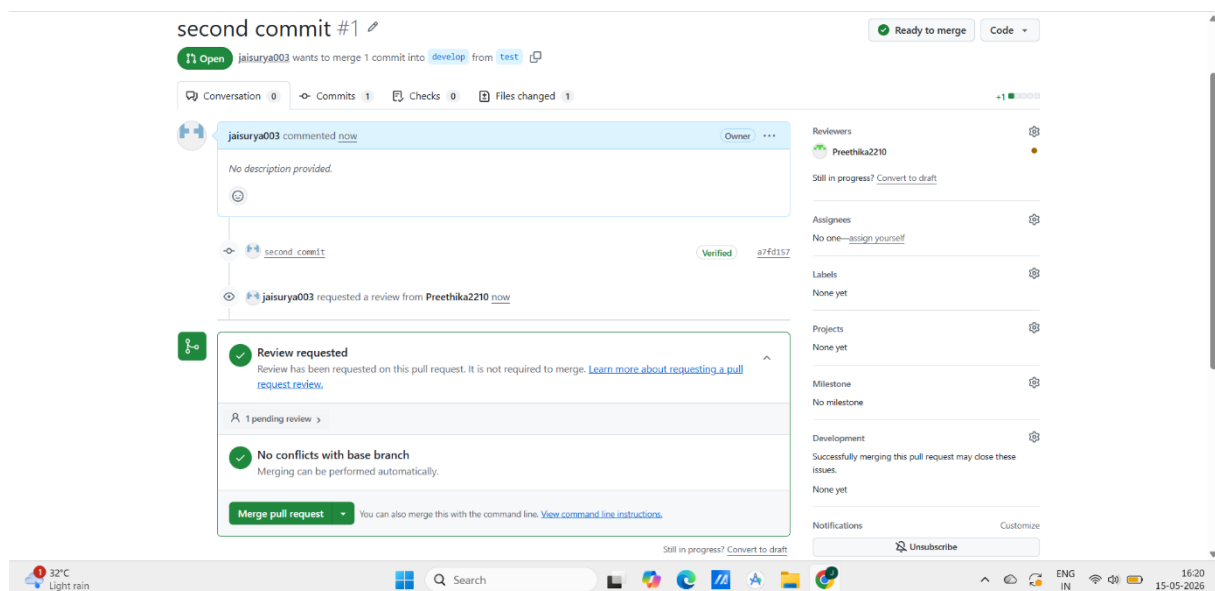
Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about comparing changes](#).



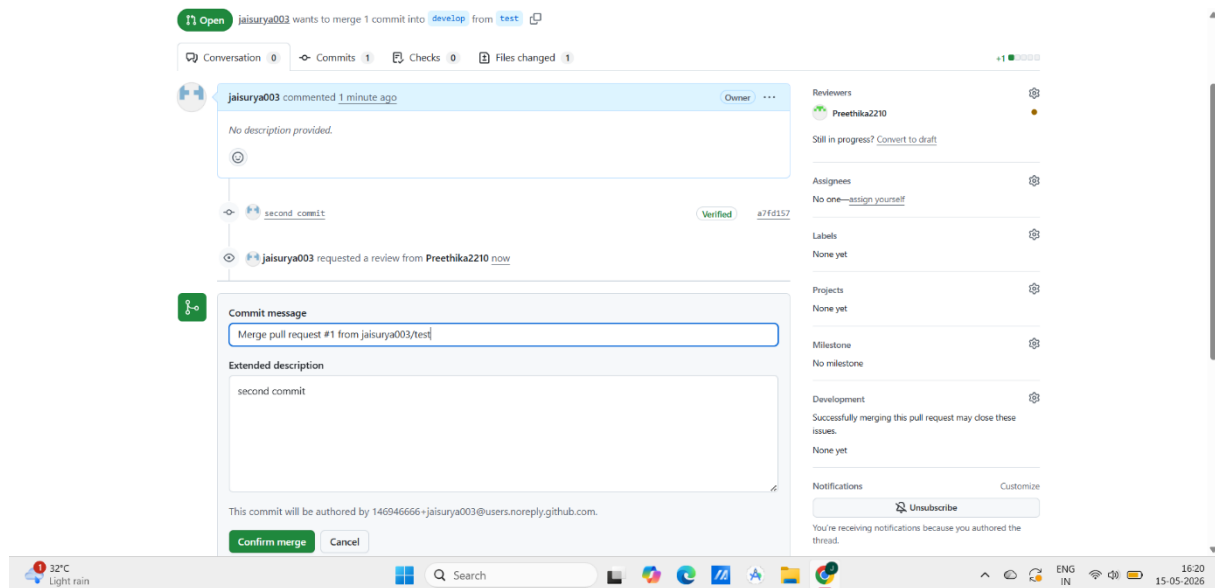
Screenshot: Merge Pull Request

- Click Merge pull request
- Merge changes from test branch into develop branch



Screenshot: Confirm Pull Request Merge

- Click Confirm merge
- Pull request merged successfully into target branch



Screenshot: Pull Request Merged Successfully

- Pull request merged successfully
- Changes updated in the target branch

